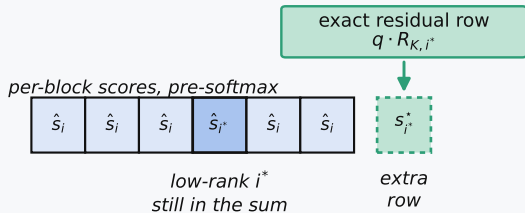


(a) residual appended as an extra softmax token

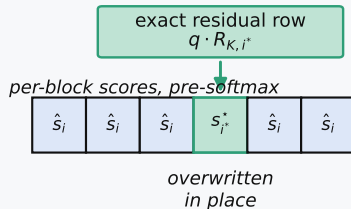


$$Z_a = \sum e^{\hat{s}_i} + e^{s_{i^*}^*}$$

$B+1$ terms — token i^* is counted twice

$Z_a > Z_b \rightarrow$ every weight in the block is pulled down by a ghost entry that no token owns

(b) residual scattered in place at res_K_positions



$$Z_b = \sum e^{s_i}, \text{ with } s_{i^*} \leftarrow s_{i^*}^*$$

B terms — token i^* counted once, exactly

denominator unchanged $\rightarrow$ the exact row corrects the score without adding weight